might convince yourself that you could. The community wasn't sure what to make of Heisenberg, but they were ready for Schrödinger. The Copenhagen crew—especially the youngsters, Heisenberg and Pauli—didn't react graciously to the competing ideas from an undistinguished old man in Zürich. But before too long they were thinking in terms of wave functions, just like everyone else.

Schrödinger's equation involves unfamiliar symbols, but its basic message is not hard to understand. De Broglie had suggested that the momentum of a wave goes up as its wavelength goes down. Schrödinger proposed a similar thing, but for energy and time: the rate at which the wave function is changing is proportional to how much energy it has. Here is the celebrated equation in its most general form:

$$\frac{\partial \Psi}{\partial t} = \frac{1}{i\hbar} H\Psi$$

We don't need the details here, but it's nice to see the real way that physicists think of an equation like this. There's some maths involved, but ultimately it's just a translation into symbols of the idea we wrote down in words.

Ψ (the Greek letter Psi) is the wave function. The left-hand side is the rate at which the wave function is changing over time. On the right-hand side we have a proportionality constant involving Planck's constant $\hbar$, the fundamental unit of quantum mechanics, and i, the square root of minus one. The wave function Ψ is acted on by something called the *Hamiltonian*, or H. Think of the Hamiltonian as an inquisitor who asks the following question: "How much energy do you have?" The concept was invented in 1833 by Irish mathematician William Rowan Hamilton, as a way to reformulate the laws of motion of a classical system, long before it gained a central role in quantum mechanics.

When physicists start modeling different physical systems, the first thing they try to do is work out a mathematical expression for the Hamiltonian of that system. The standard way of figuring out the Hamiltonian of something like a collection of particles is to start with the energies of the particles themselves, and then add in additional contributions describing how the particles interact with each other. Maybe they bump off each other like billiard balls, or perhaps they exert a mutual